

Central Limit Theorem

Central Limit Theorem, Normality, and Sampling Distributions

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Some types of samples:

- **Probability:** known probability of selection
- **Nonprobability:** unknown probability of selection
- **Random:** equal chance of being selected
- **Systematic:** systematic selection (e.g. every 9th element)
- **Stratified:** selection from strata to match population demographics

Examples of sample biases:

- **Undercoverage bias:** e.g., phone surveys require phones
- **Self-selection bias:** e.g., some people are more likely to take surveys
- **Survivorship bias:** e.g., asking current students misses non-applicants
- **Non-response bias:** e.g., are people who not responding for a reason?
- **Recall bias:** e.g., people can mis-remember, satisfice, or lie
- **Observer bias:** e.g., tone may signal “better” or “worse” answers

Avoiding/Mitigating Sampling Bias

DO's:

- Use random probability samples.
- Use stratified sampling or weighting.
- Follow up with non-responders for surveys.
- Acknowledge, investigate, and mitigate missing values.
- Oversample certain populations to ensure representation, then stratify/weight.

DON'Ts:

- DON'T Use convenience sampling
- DON'T Deviate too much from the 95% confidence interval and a 3-6% margin of error in calculating sample size.
- DON'T Limit to one sampling mode (e.g., phone, email, web, etc.)
- DON'T Over-complicate tasks or overburden participants

So What?

Researchers need to be clear and transparent about sampling:

1. The method of selection (how observations were collected)
2. The number of observations (how many cases the sample contains)

Good samples are:

1. Random
2. Representative
3. Big enough

We can avoid lots of sampling biases by combining **random** sampling, **probability** sampling, and **stratified** sampling techniques.

When done right, sampling can be **more reliable** than directly polling the population, and is certainly more cost effective.

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Remember: a goal of social science research is to make *generalizable* inferences about the population from samples.

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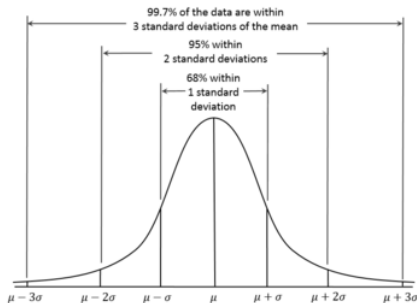
In other words: the whole reason we can make inferences about the population from good samples is because math says that enough sample means will eventually form a normal distribution centered around the population mean!

Central Limit Theorem

The standard normal distribution is the backbone of statistical inference.

The Central Limit Theorem also means that 95% of the observations will fall within 2 standard deviations of the mean of a normal distribution.

- This is where we get the 95% confidence interval...
- ...and the norm of significance for p-values of less than 0.05



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Coin Tossing

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With enough samples, the histogram of each sample's mean will be distributed normally, centered around the mean of .5

Practice Sampling Distributions

Range: 1–100; Sample size: 12 Obs.; # Samples: 10

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
Ob 1	6	96	85	70	30	59	65	38	71	97
Ob 2	11	48	81	55	59	79	39	95	34	49
Ob 3	25	61	44	5	77	74	69	97	98	41
Ob 4	42	98	81	83	95	72	61	30	77	28
Ob 5	5	7	75	52	21	38	66	76	99	84
Ob 6	8	69	91	22	44	9	29	54	19	68
Ob 7	63	73	51	22	92	78	10	33	12	1
Ob 8	30	24	29	23	52	90	29	98	44	3
Ob 9	32	34	43	87	52	47	92	67	2	23
Ob 10	49	89	64	23	1	28	81	38	72	35
Ob 11	94	48	21	29	75	9	69	44	99	28
Ob 12	1	88	67	84	81	69	9	43	100	7